EDA and MCMC Methods for Babies Dataset

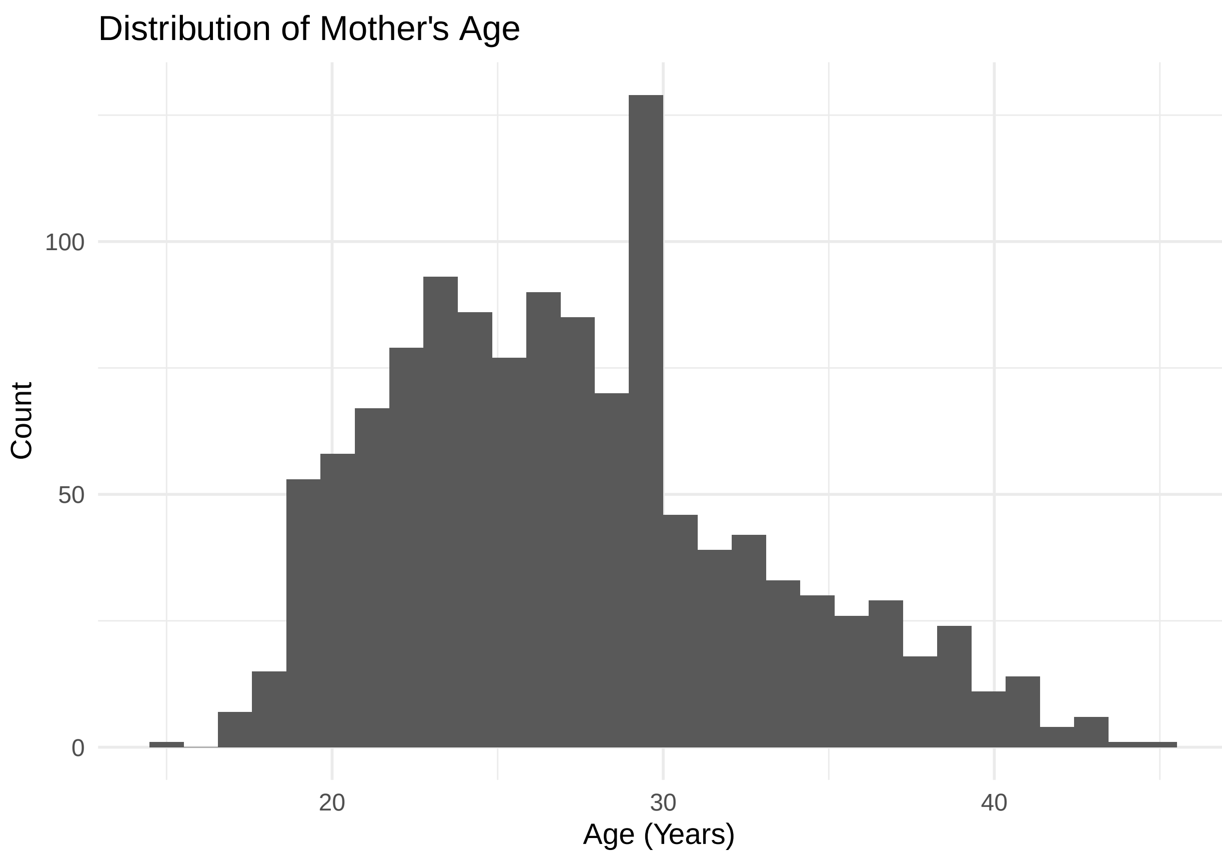
Yatziri I. Carmona Ochoa

2026-04-26

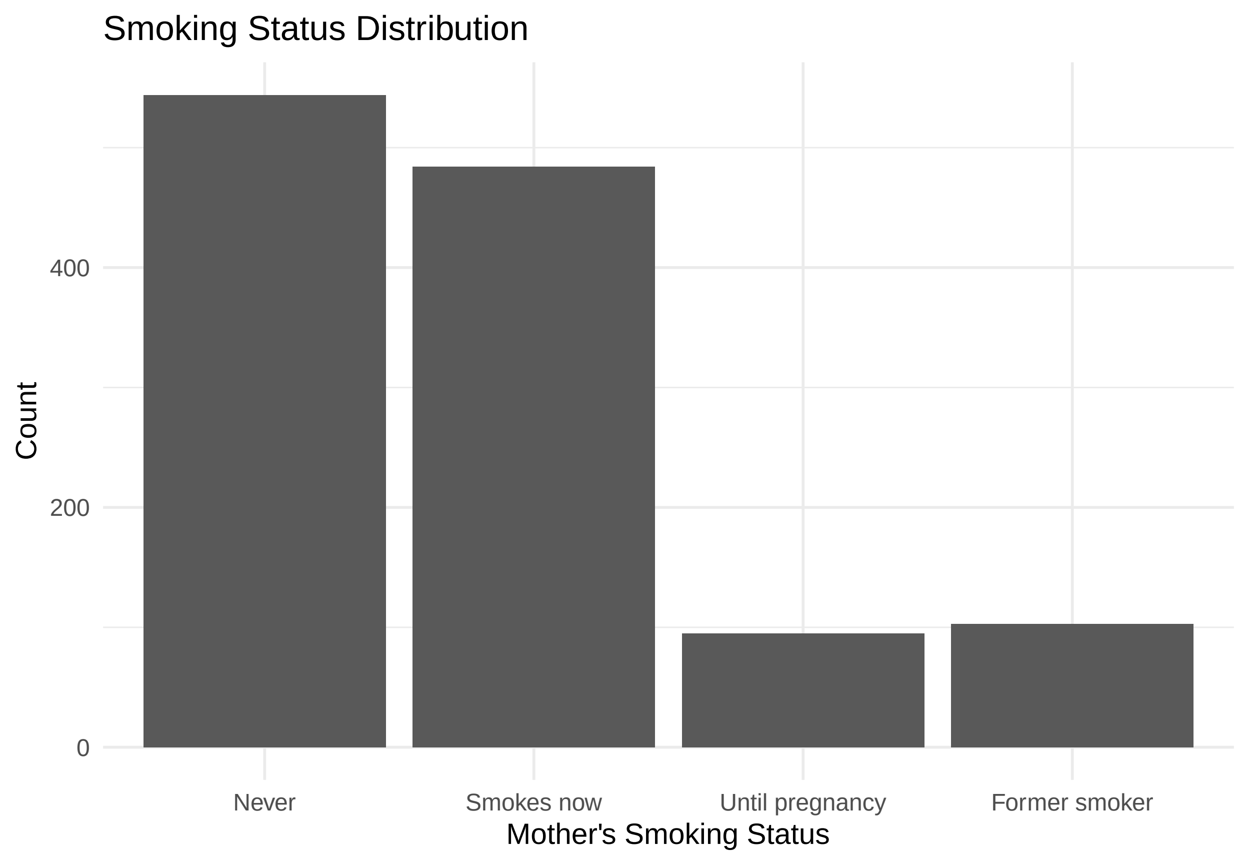
```r
library(UsingR)
library(Bolstad)
library(ggplot2)
library(reshape2)
library(ggridges)
```

EDA

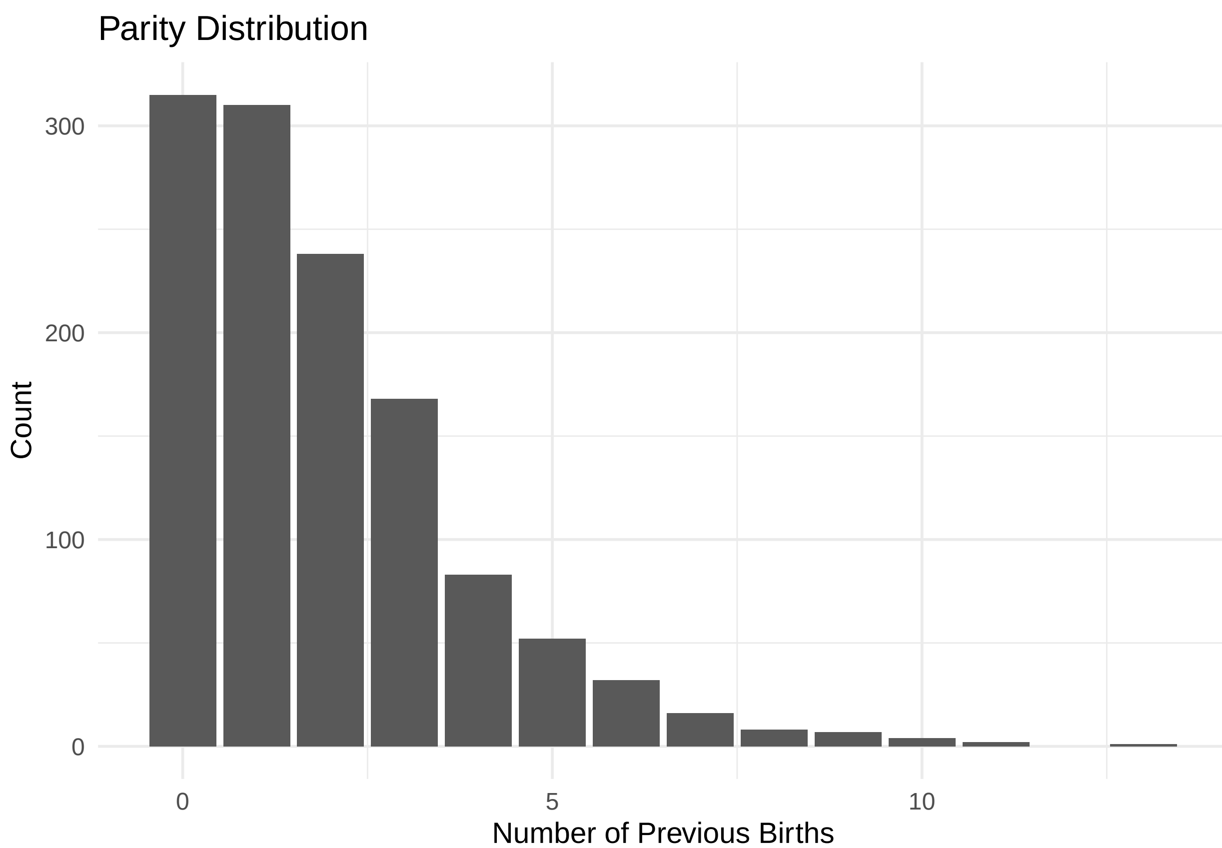
```r
babies_clean <- babies

#removing missing/incomplete data
babies_clean[babies_clean == 999] <- NA
babies_clean$age[babies_clean$age == 99] <- NA
babies_clean$wt1[babies_clean$wt1 == 999] <- NA
babies_clean$smoke[babies_clean$smoke == 9] <- NA
babies_clean$race[babies_clean$race == 99] <- NA
babies_clean$inc[babies_clean$inc == 99] <- NA
babies_clean$inc[babies_clean$inc == 98] <- NA
babies_clean$inc[babies_clean$sex == 9] <- NA
babies_clean$inc[babies_clean$ed == 9] <- NA
babies_clean$inc[babies_clean$ht == 99] <- NA

#age distribution
ggplot(babies_clean, aes(x = age)) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of Mother's Age",
       x = "Age (Years) ", y = "Count") +
  theme_minimal()
```



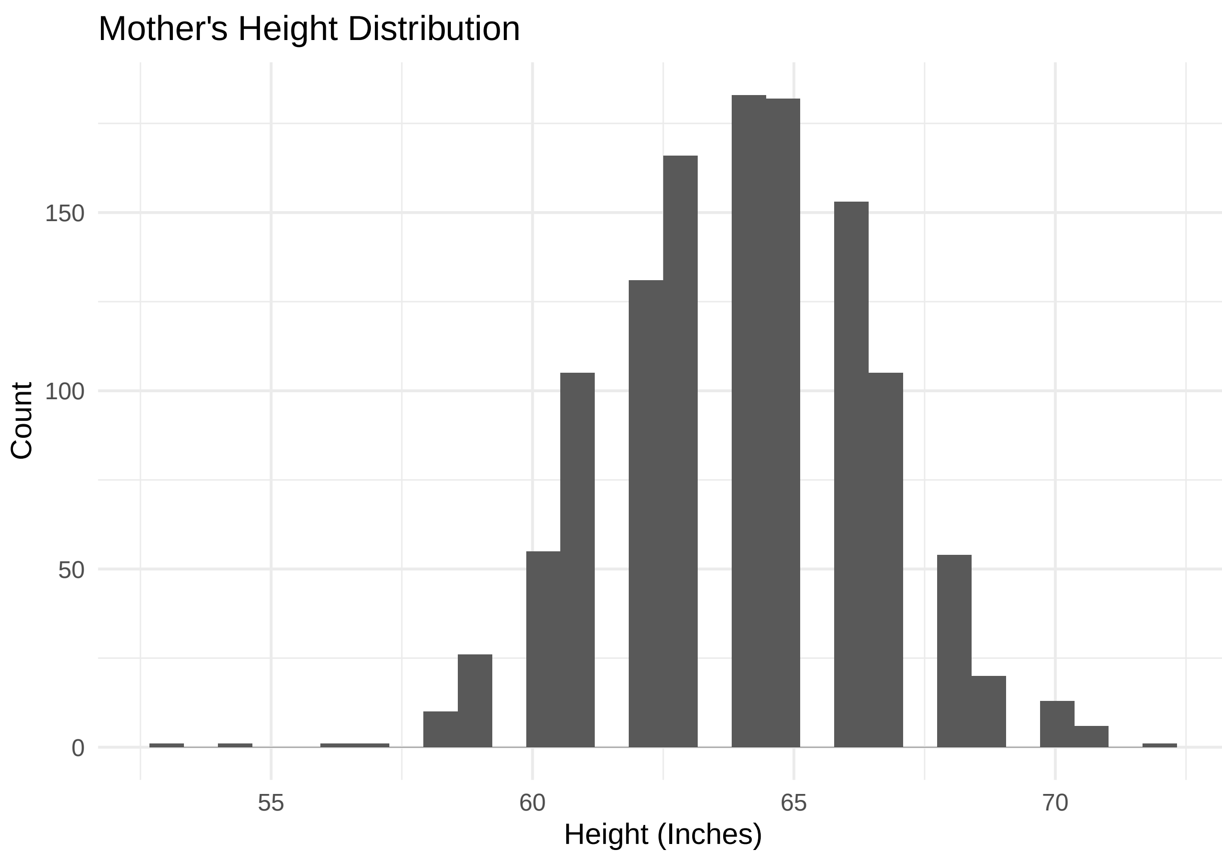

```
#smoking status distributions
babies_clean$smoke_group <- with(babies_clean,
  ifelse(smoke == 0, "Never",
  ifelse(smoke == 1, "Smokes now",
  ifelse(smoke == 2, "Until pregnancy",
  ifelse(smoke == 3, "Former smoker", NA)))))

babies_clean$smoke_group <- factor(
  babies_clean$smoke_group,
  levels = c("Never", "Smokes now", "Until pregnancy", "Former smoker")
)

ggplot(subset(babies_clean, !is.na(smoke_group)), aes(x = smoke_group)) +
  geom_bar() +
  labs(
    title = "Smoking Status Distribution",
    x = "Mother's Smoking Status",
    y = "Count"
  ) +
  theme_minimal()
```



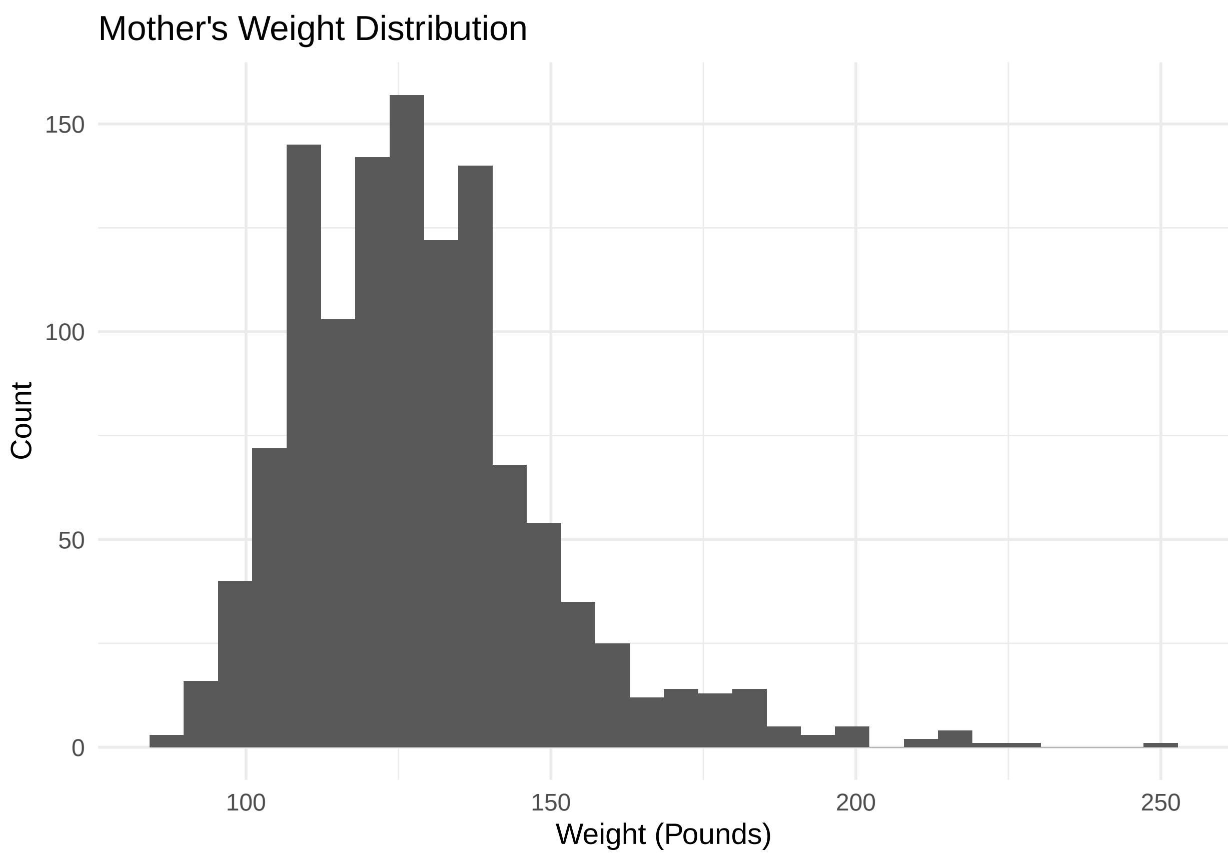

```
#parity distribution
ggplot(babies_clean, aes(x = parity)) +
  geom_bar() +
  labs(title = "Parity Distribution",
       x = "Number of Previous Births", y = "Count") +
  theme_minimal()
```



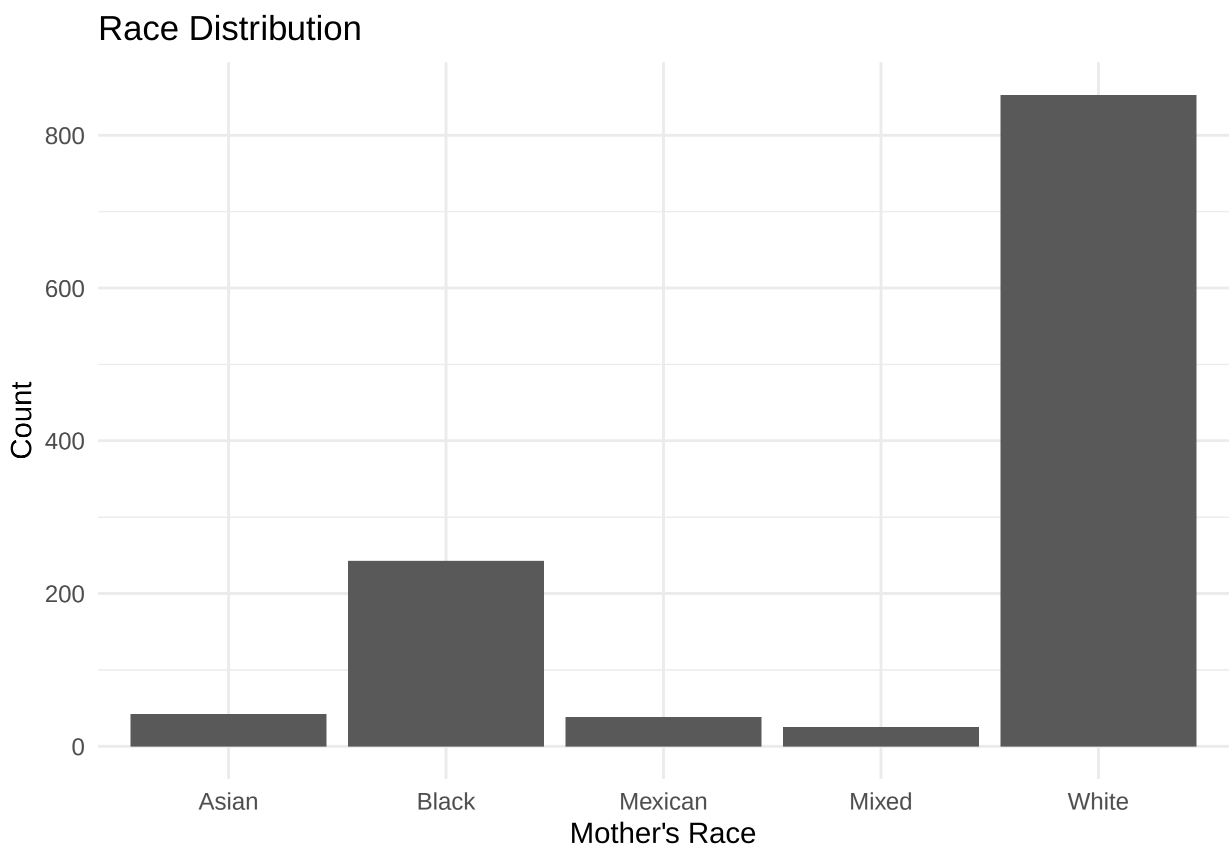

```r
#removing height outlier
babies_clean <- babies_clean[babies_clean$ht != 99, ]

#height distribution for moms
ggplot(babies_clean, aes(x = ht)) +
  geom_histogram(bins = 30) +
  labs(title = "Mother's Height Distribution",
       x = "Height (Inches)", y = "Count") +
  theme_minimal()
```




```
#mother's weight distribution
ggplot(babies_clean, aes(x = wt1)) +
  geom_histogram(bins = 30) +
  labs(title = "Mother's Weight Distribution",
       x = "Weight (Pounds)", y = "Count") +
  theme_minimal()
```

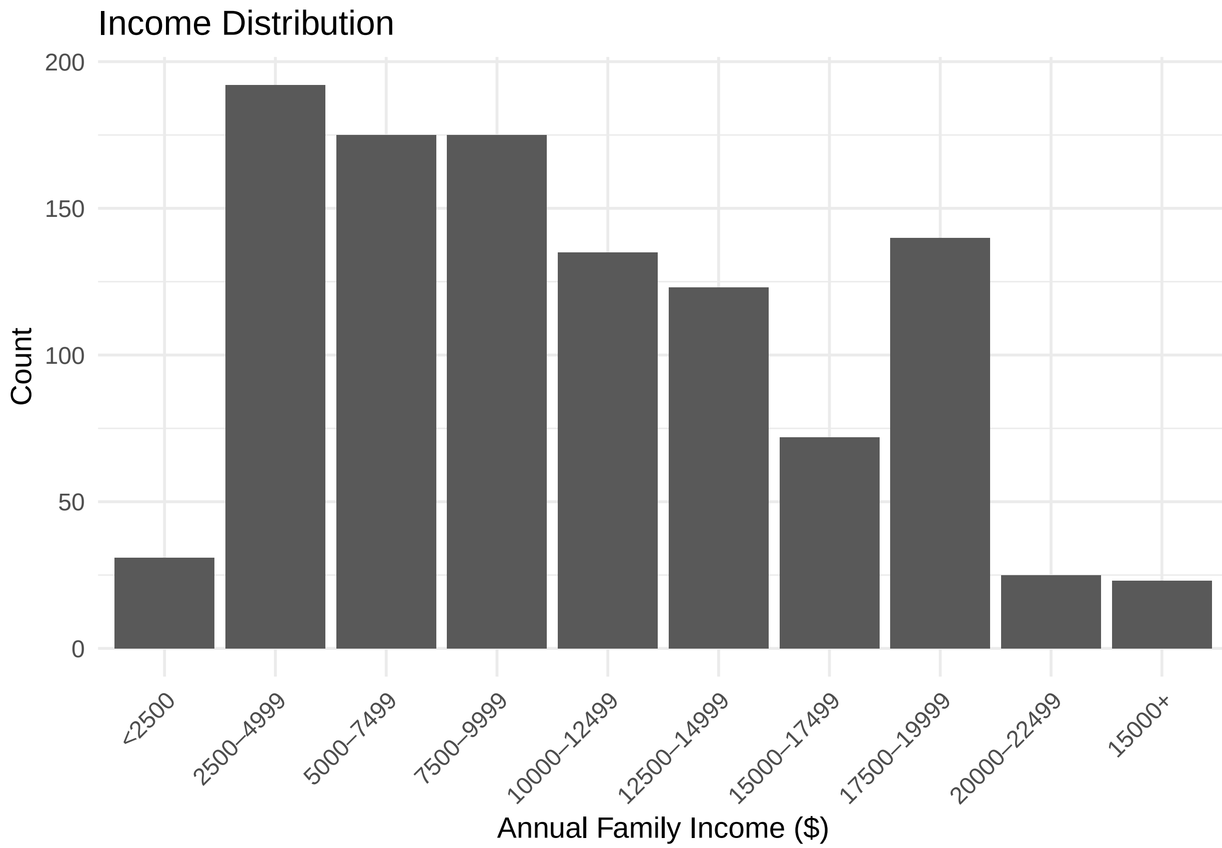



```r
#mother's race distribution
babies_clean$race_group <- with(babies_clean,
  ifelse(race %in% 0:5, "White",
  ifelse(race == 6, "Mexican",
  ifelse(race == 7, "Black",
  ifelse(race == 8, "Asian",
  ifelse(race == 9, "Mixed", NA))))))

babies_clean$race_group[babies_clean$race == 10] <- NA

ggplot(subset(babies_clean, !is.na(race_group)), aes(x = race_group)) +
  geom_bar() +
  labs(
    title = "Race Distribution",
    x = "Mother's Race",
    y = "Count"
  ) +
  theme_minimal()
```

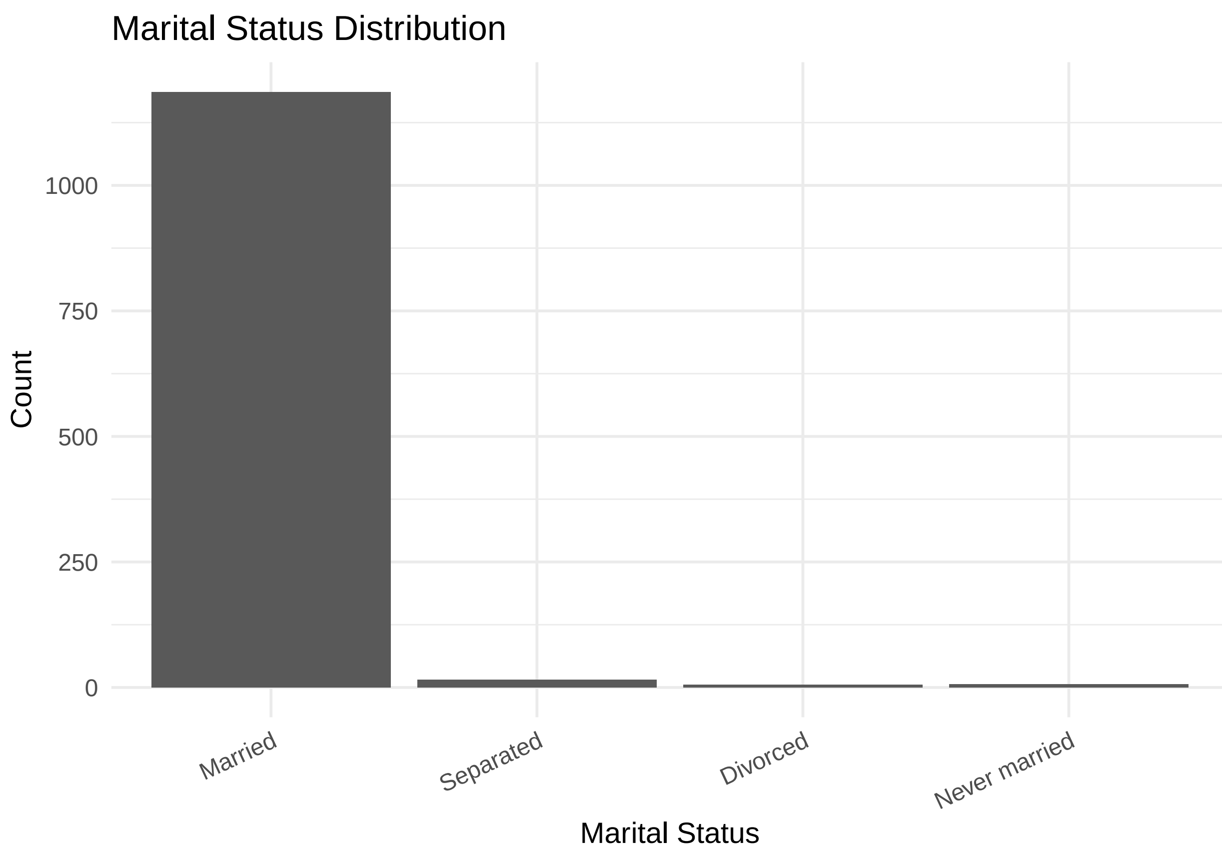



```r
#annual income
babies_clean$inc_group <- factor(
  babies_clean$inc,
  levels = 0:9,
  labels = c(
    "<2500",
    "2500-4999",
    "5000-7499",
    "7500-9999",
    "10000-12499",
    "12500-14999",
    "15000-17499",
    "17500-19999",
    "20000-22499",
    "15000+"
  )
)

ggplot(subset(babies_clean, !is.na(inc_group)), aes(x = inc_group)) +
  geom_bar() +
  labs(
    title = "Income Distribution",
    x = "Annual Family Income ($)",
    y = "Count"
  ) +
  theme_minimal() +
```

```
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

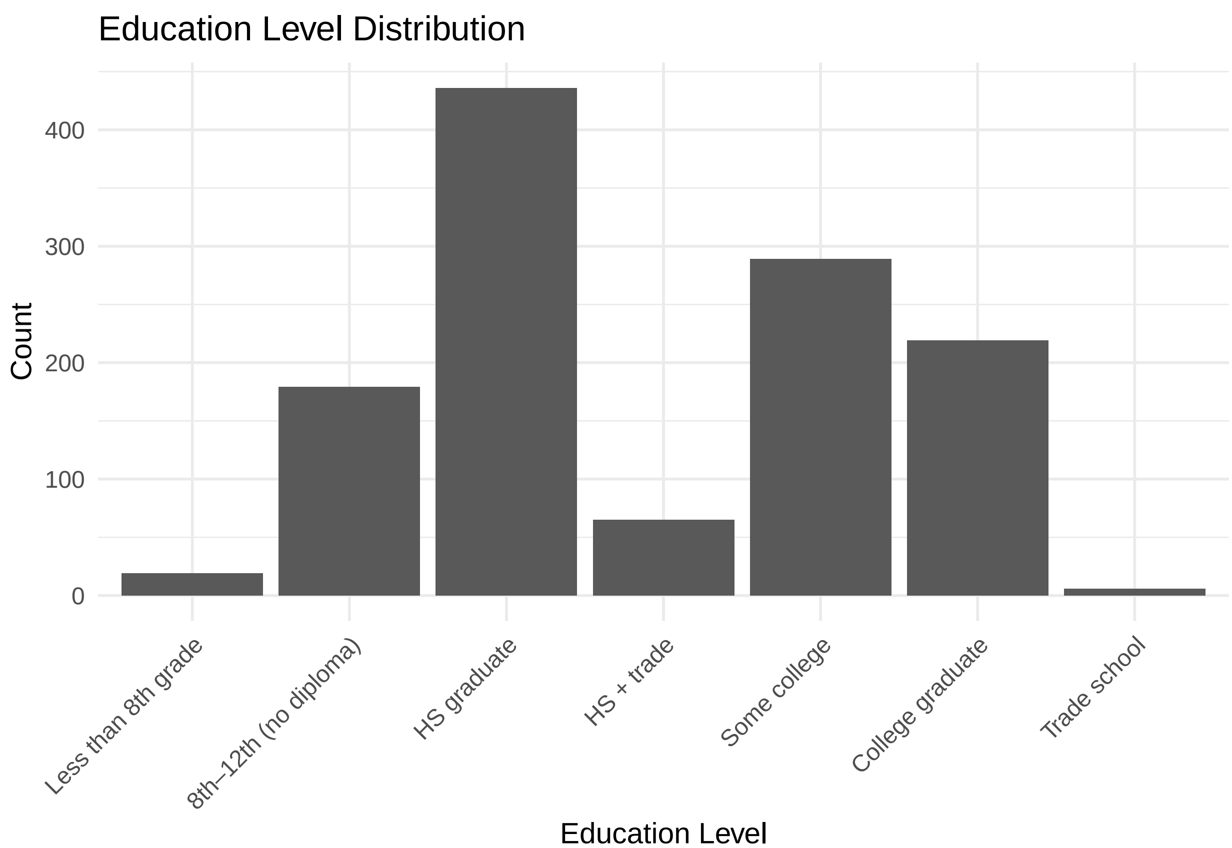



```
#marital status distribution
babies_clean$marital_group <- factor(
  babies_clean$marital,
  levels = c(1,2,3,4,5),
  labels = c(
    "Married",
    "Separated",
    "Divorced",
    "Widowed",
    "Never married"
  )
)

ggplot(subset(babies_clean, !is.na(marital_group)), aes(x = marital_group)) +
  geom_bar() +
  labs(
    title = "Marital Status Distribution",
    x = "Marital Status",
    y = "Count"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 25, hjust = 1))
```




```r
#mother's education distribution
babies_clean$edu_group <- with(babies_clean,
  ifelse(ed == 0, "Less than 8th grade",
  ifelse(ed == 1, "8th-12th (no diploma)",
  ifelse(ed == 2, "HS graduate",
  ifelse(ed == 3, "HS + trade",
  ifelse(ed == 4, "Some college",
  ifelse(ed == 5, "College graduate",
  ifelse(ed %in% c(6,7), "Trade school", NA))))))))

babies_clean$edu_group <- factor(
  babies_clean$edu_group,
  levels = c(
    "Less than 8th grade",
    "8th-12th (no diploma)",
    "HS graduate",
    "HS + trade",
    "Some college",
    "College graduate",
    "Trade school"
  )
)

ggplot(subset(babies_clean, !is.na(edu_group)), aes(x = edu_group)) +
  geom_bar() +
  labs(
```

```r
    title = "Education Level Distribution",
    x = "Education Level",
    y = "Count"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



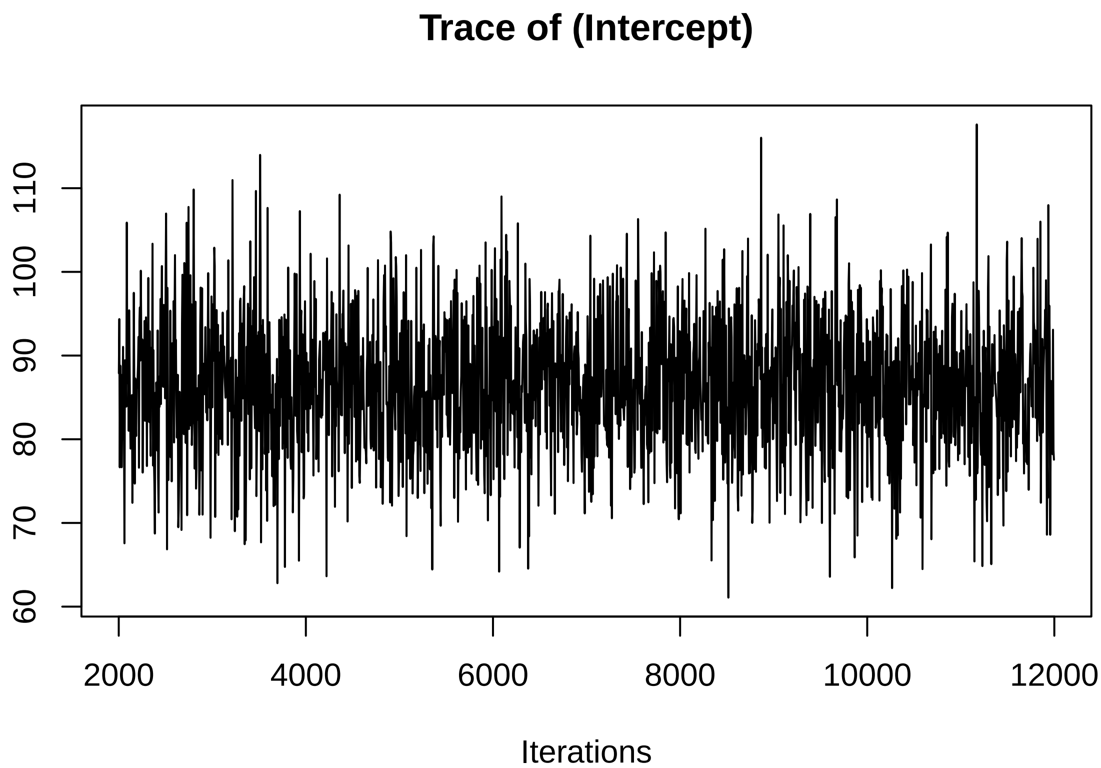
MCMC for Bayesian Linear Regression

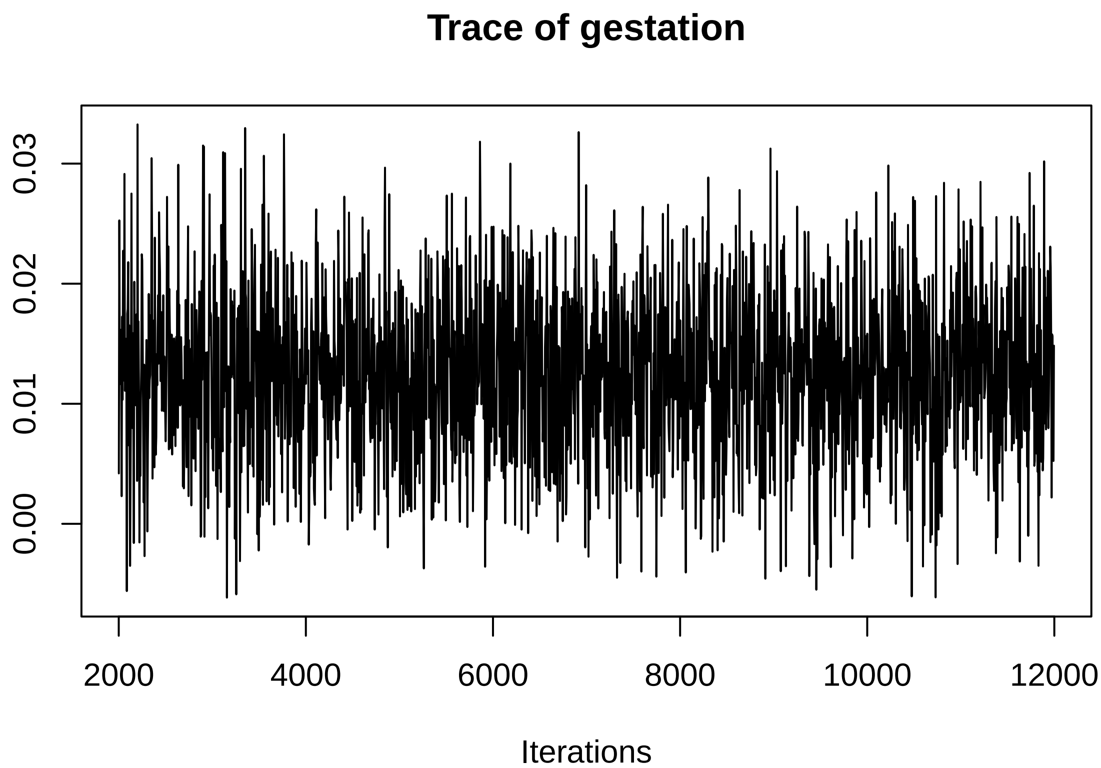
```r
library(MCMCpack)

#linear regression
model <- MCMCregress(
  wt ~ gestation + age + ht + wt1 + smoke + race + parity,
  data = babies,
  burnin = 2000,
  mcmc = 10000,
  thin = 5
)

summary(model)
```

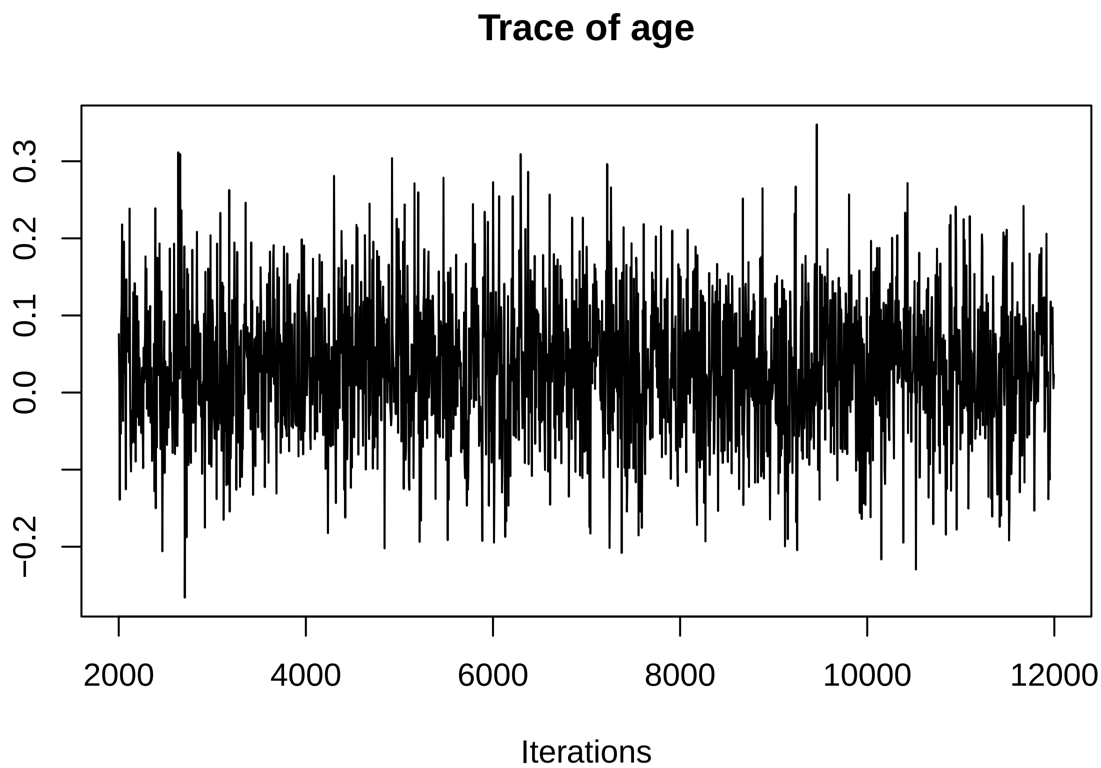
```
##
```

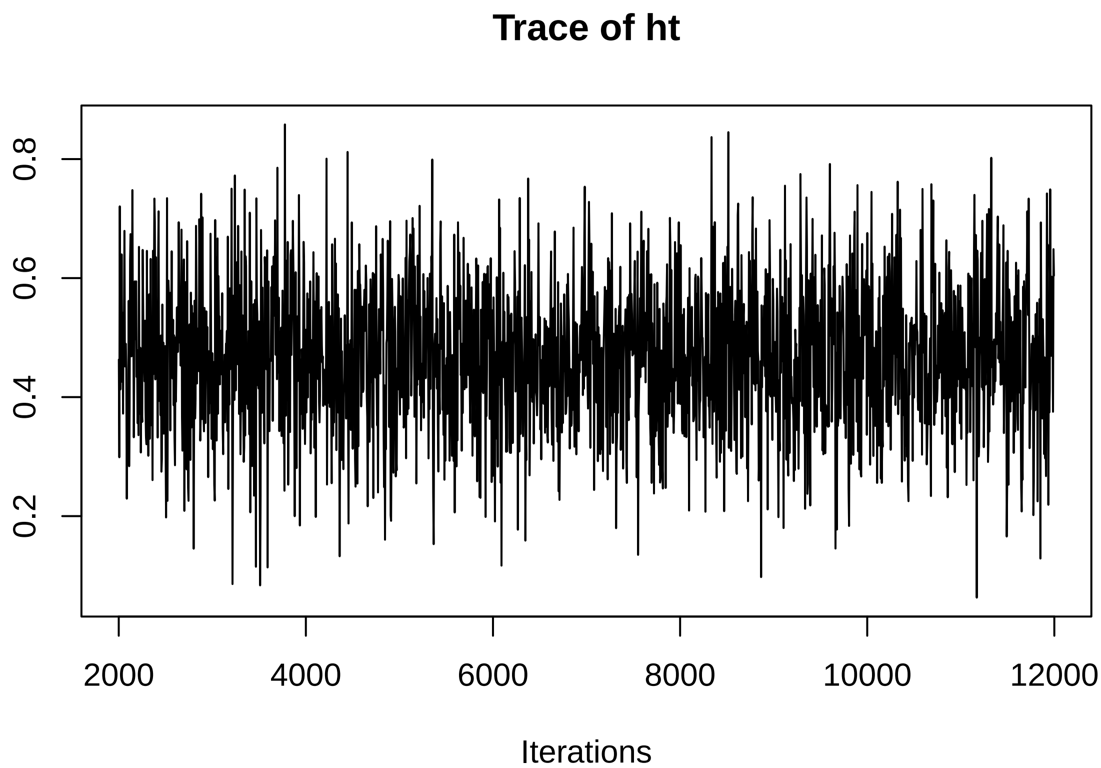
```
## Iterations = 2001:11996
## Thinning interval = 5
## Number of chains = 1
## Sample size per chain = 2000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##                   Mean        SD  Naive SE Time-series SE
## (Intercept)  86.519359  8.049037 1.800e-01      0.1799819
## gestation     0.012502  0.006873 1.537e-04      0.0001427
## age           0.031569  0.090759 2.029e-03      0.0020294
## ht            0.466948  0.122381 2.737e-03      0.0027365
## wt1          -0.004533  0.004356 9.741e-05      0.0000930
## smoke        -0.074574  0.439143 9.820e-03      0.0101861
## race         -0.447499  0.127816 2.858e-03      0.0025491
## parity        0.309394  0.297464 6.651e-03      0.0061220
## sigma2      323.783633 13.274759 2.968e-01      0.2968326
##
## 2. Quantiles for each variable:
##
##                  2.5%        25%      50%        75%      97.5%
## (Intercept)  70.62800  81.151355 86.46601  91.696259 102.173169
## gestation    -0.00100   0.007903  0.01251   0.017023   0.025836
## age          -0.14676  -0.029559  0.02978   0.092347   0.212094
## ht            0.22728   0.384350  0.46822   0.549919   0.706145
## wt1          -0.01302  -0.007545 -0.00451  -0.001594   0.003933
## smoke        -0.93418  -0.369793 -0.07249   0.207826   0.834490
## race         -0.69789  -0.532310 -0.44748  -0.360221  -0.204593
## parity       -0.28695   0.111771  0.30776   0.504304   0.922877
## sigma2      298.86122 315.010496 323.56163 332.388106 350.258303
```

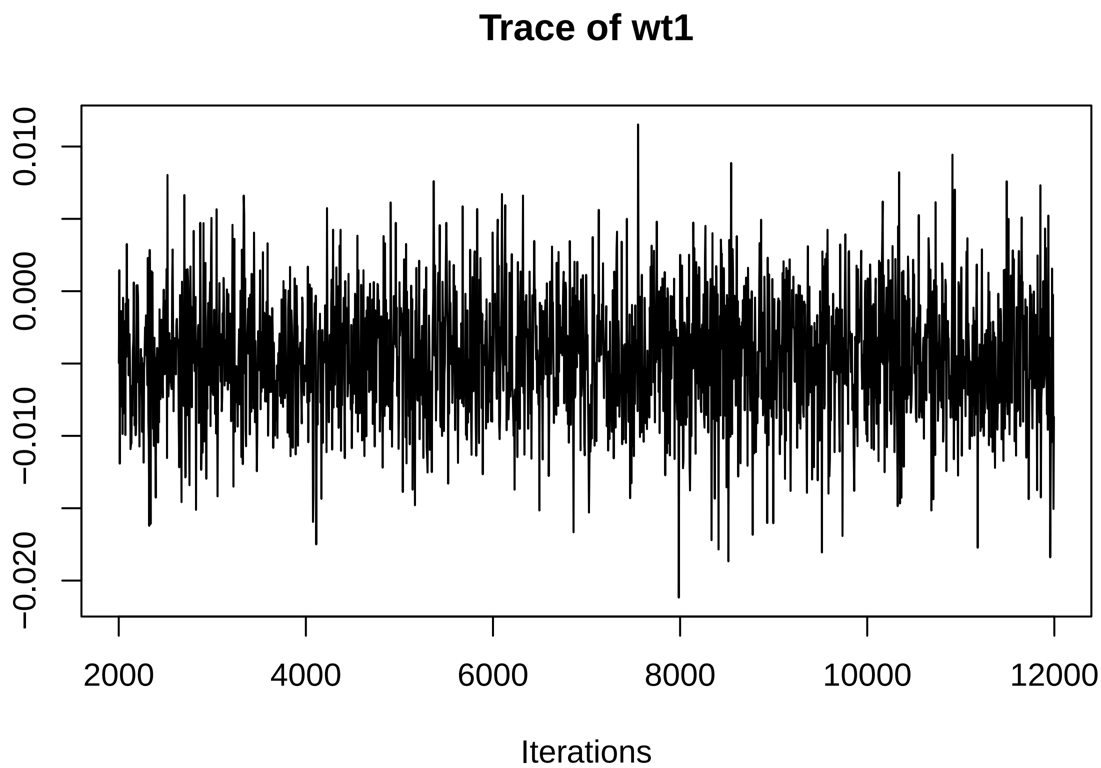
```r
#diagnostics
traceplot(model)
```

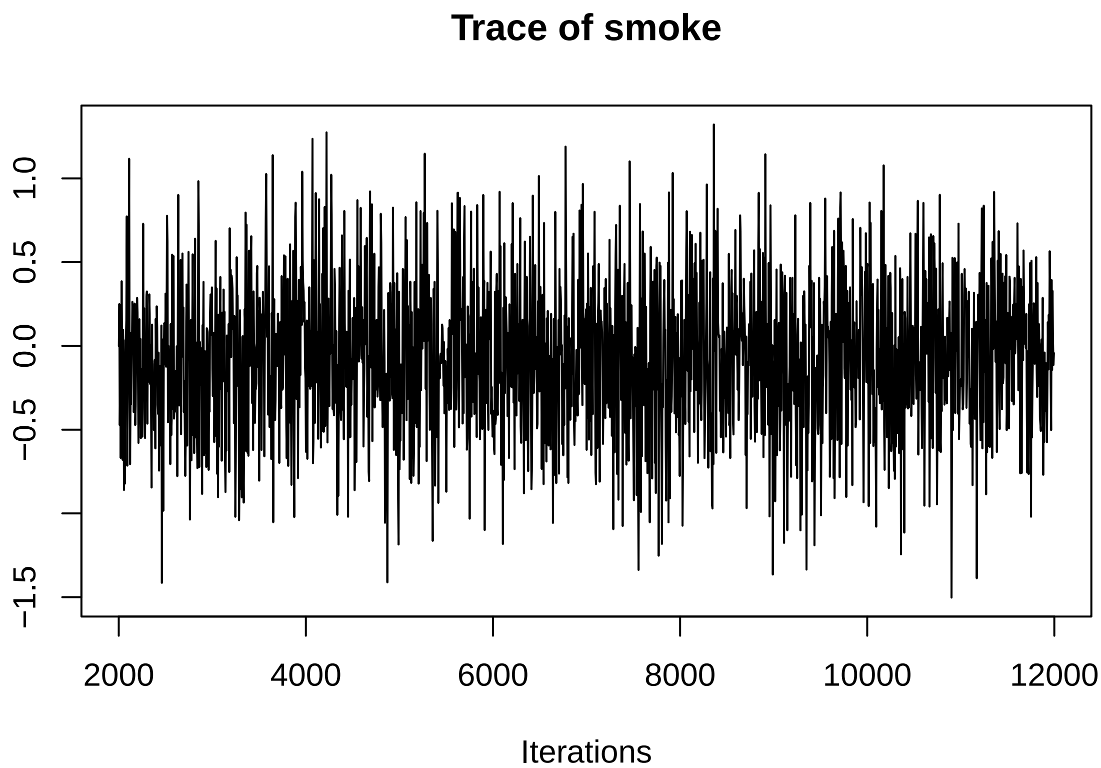

Trace of (Intercept)
60
70
80
90
100
110
2000
4000
6000
8000
10000
12000
Iterations

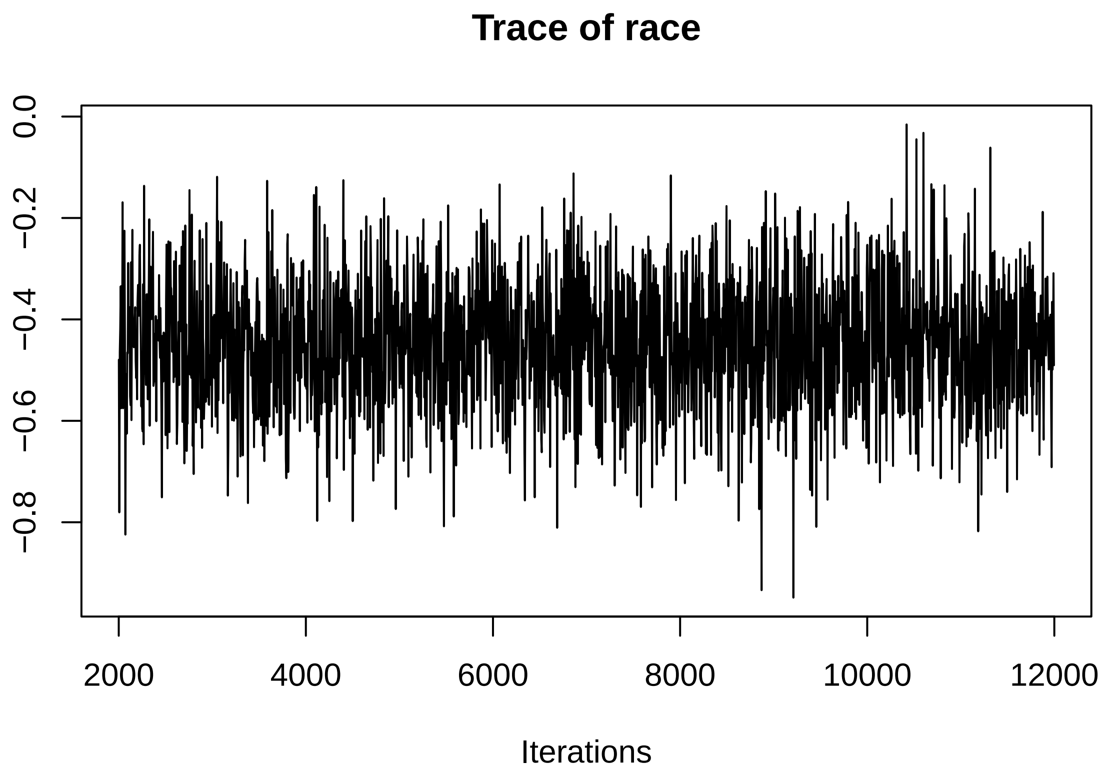

Trace of gestation
0.00
0.01
0.02
0.03
2000
4000
6000
8000
10000
12000
Iterations


Trace of age
0.3
0.2
0.1
0.0
−0.2
2000
4000
6000
8000
10000
12000
Iterations

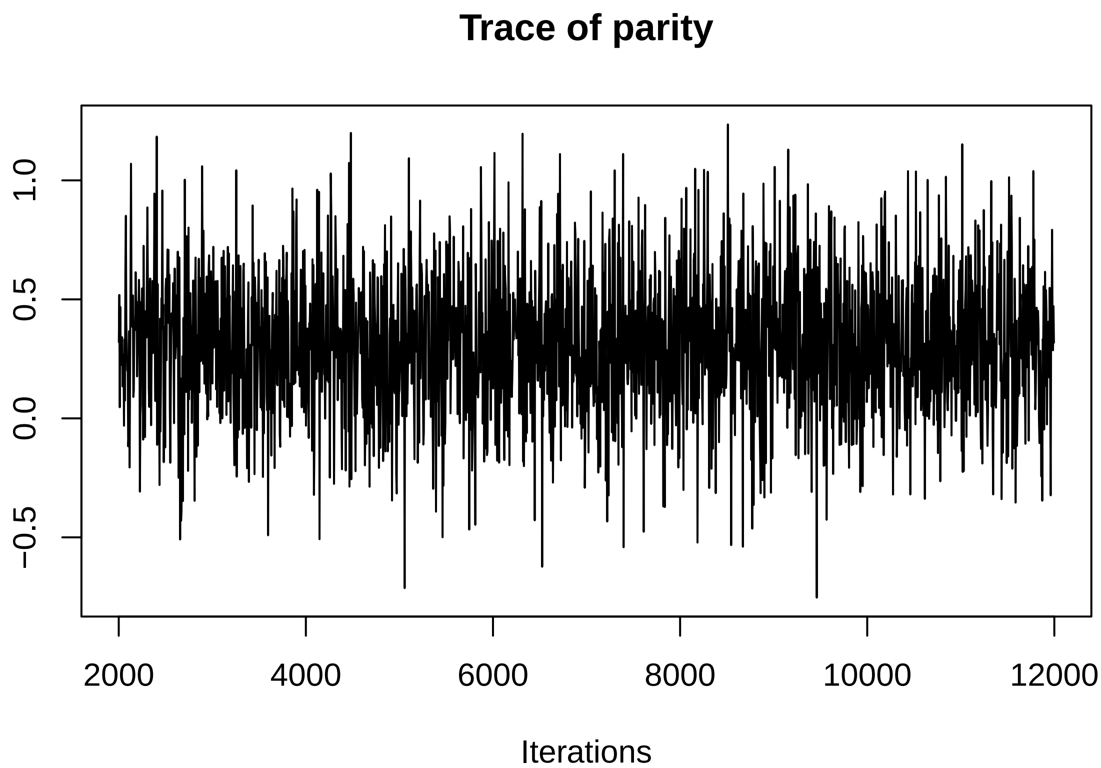

Trace of ht
0.2
0.4
0.6
0.8
2000
4000
6000
8000
10000
12000
Iterations

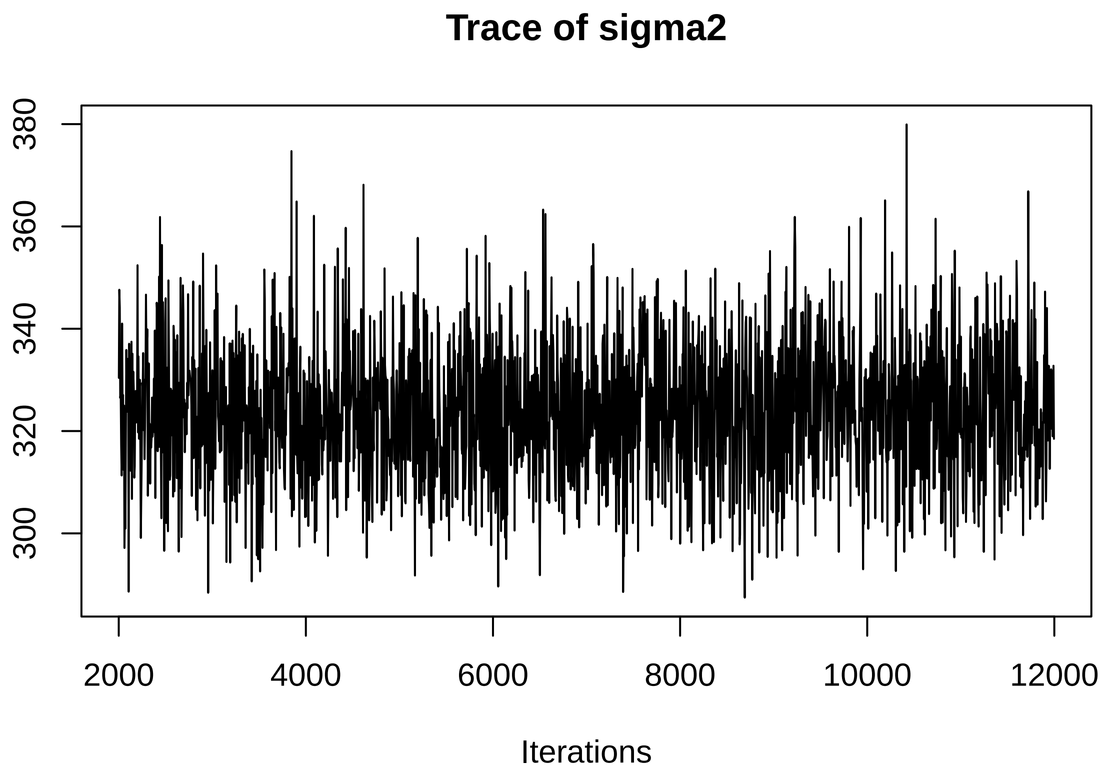

Trace of wt1
0.010
0.000
−0.010
−0.020
2000
4000
6000
8000
10000
12000
Iterations


Trace of smoke
1.0
0.5
0.0
−0.5
−1.5
2000
4000
6000
8000
10000
12000
Iterations

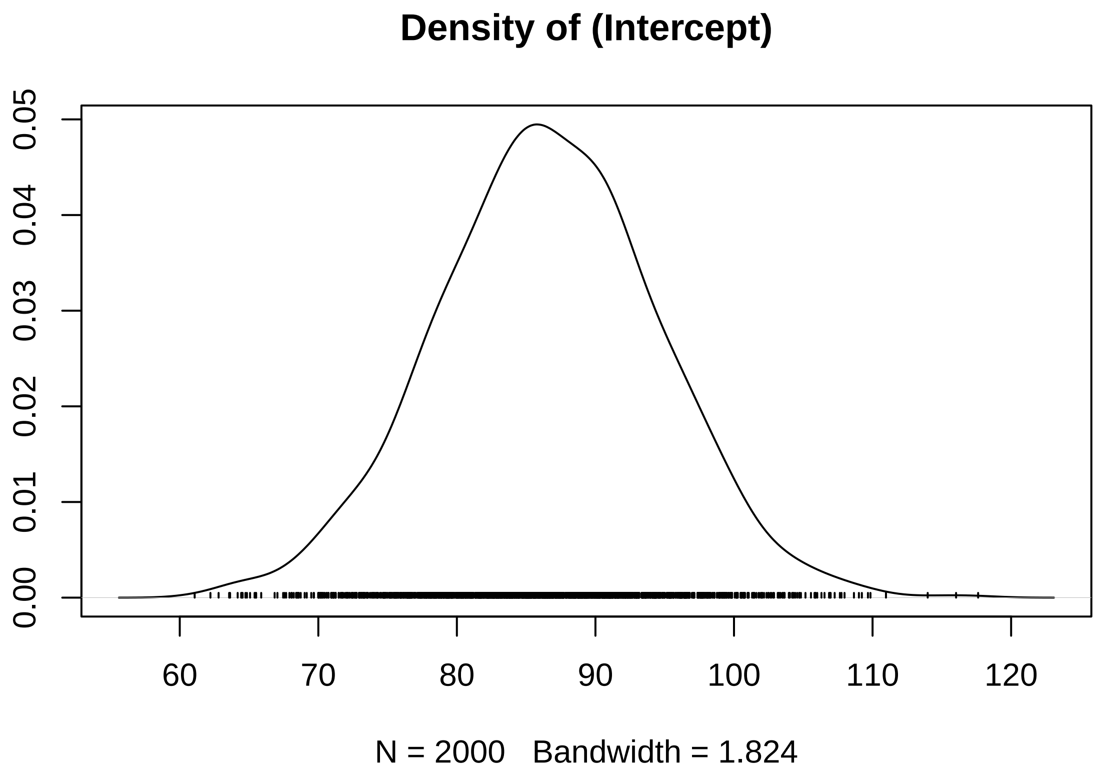


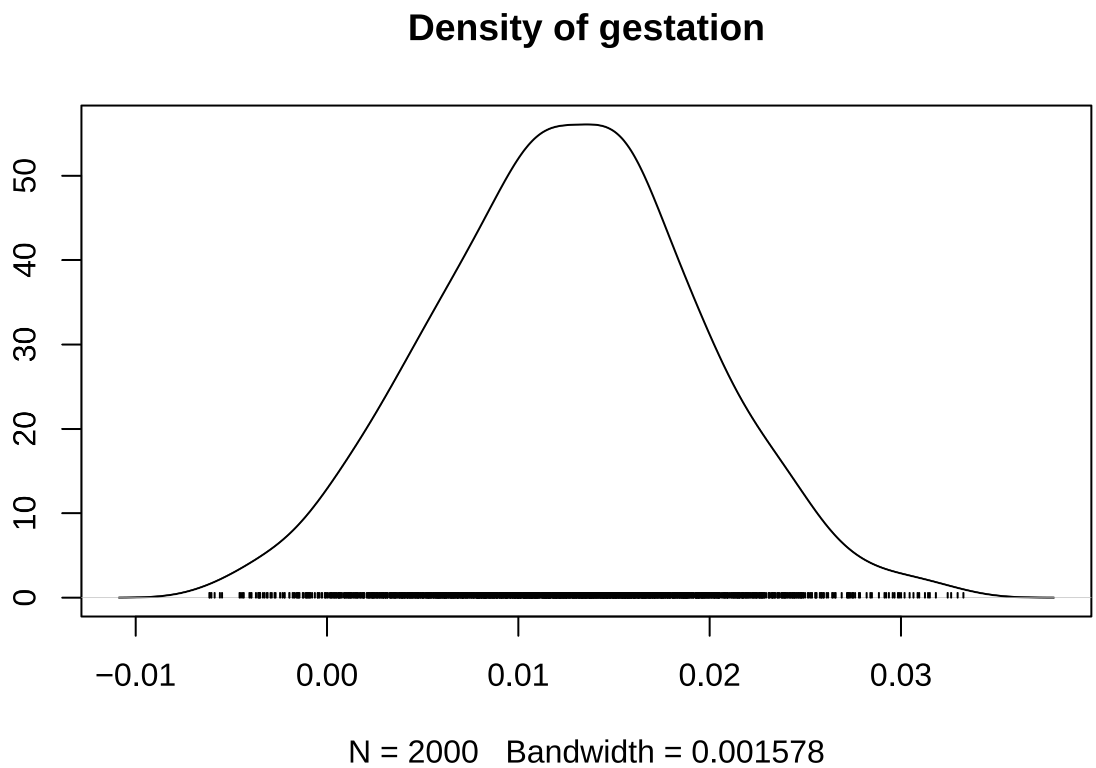
Trace of race
0.0
−0.2
−0.4
−0.6
−0.8
2000
4000
6000
8000
10000
12000
Iterations

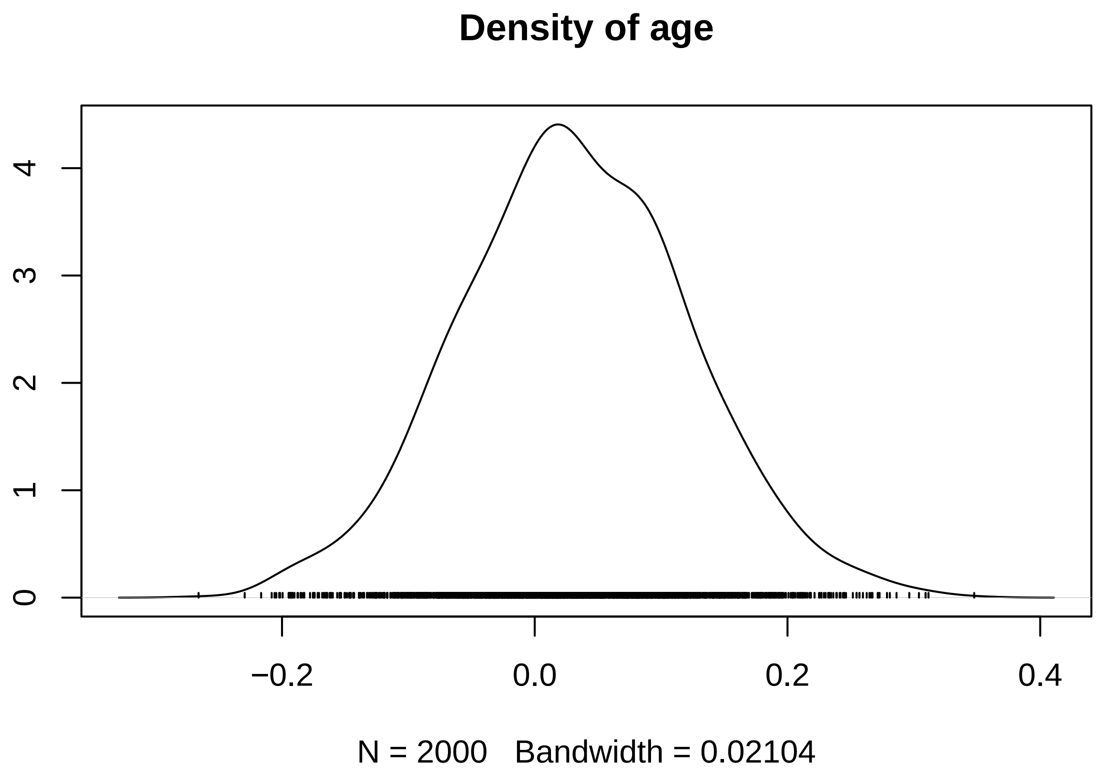

Trace of parity
1.0
0.5
0.0
−0.5
2000
4000
6000
8000
10000
12000
Iterations

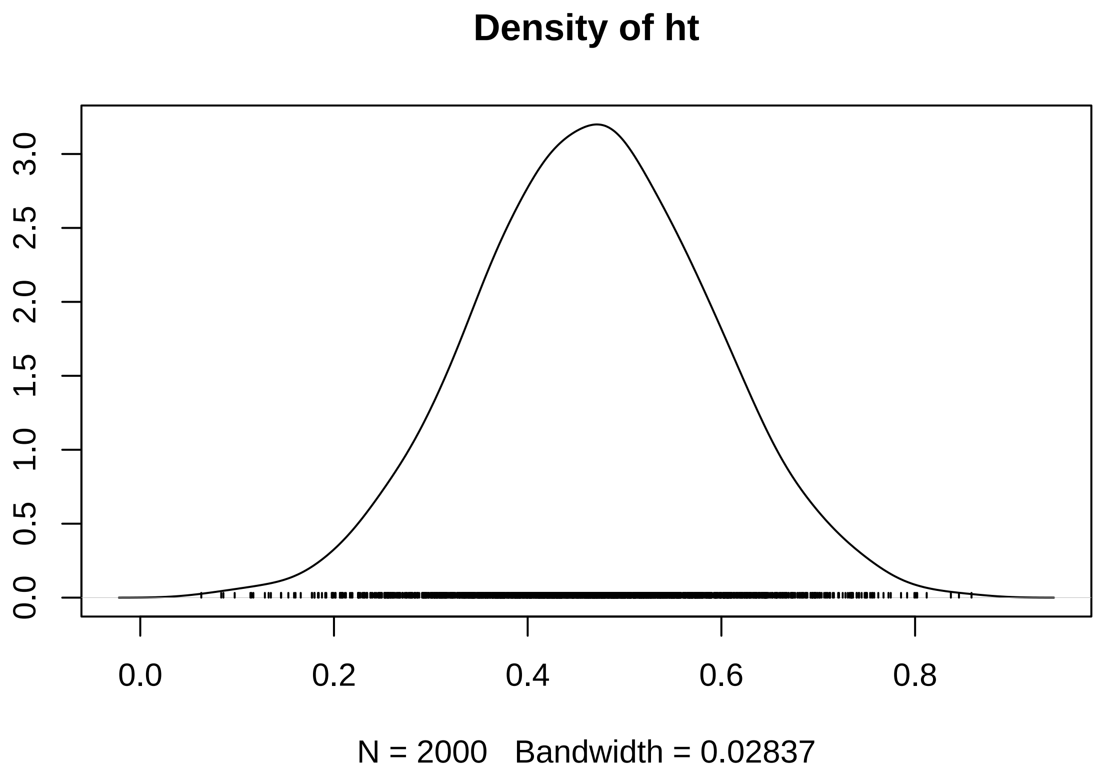


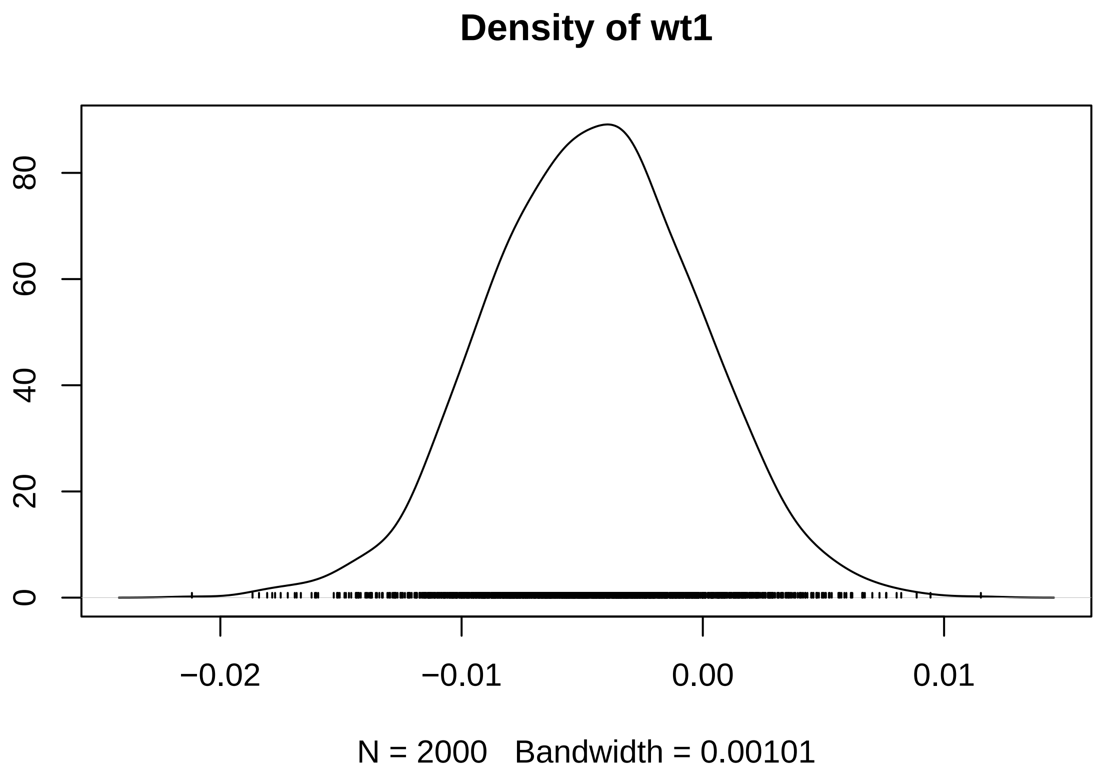

```
#posterior distributions
densplot(model)
```

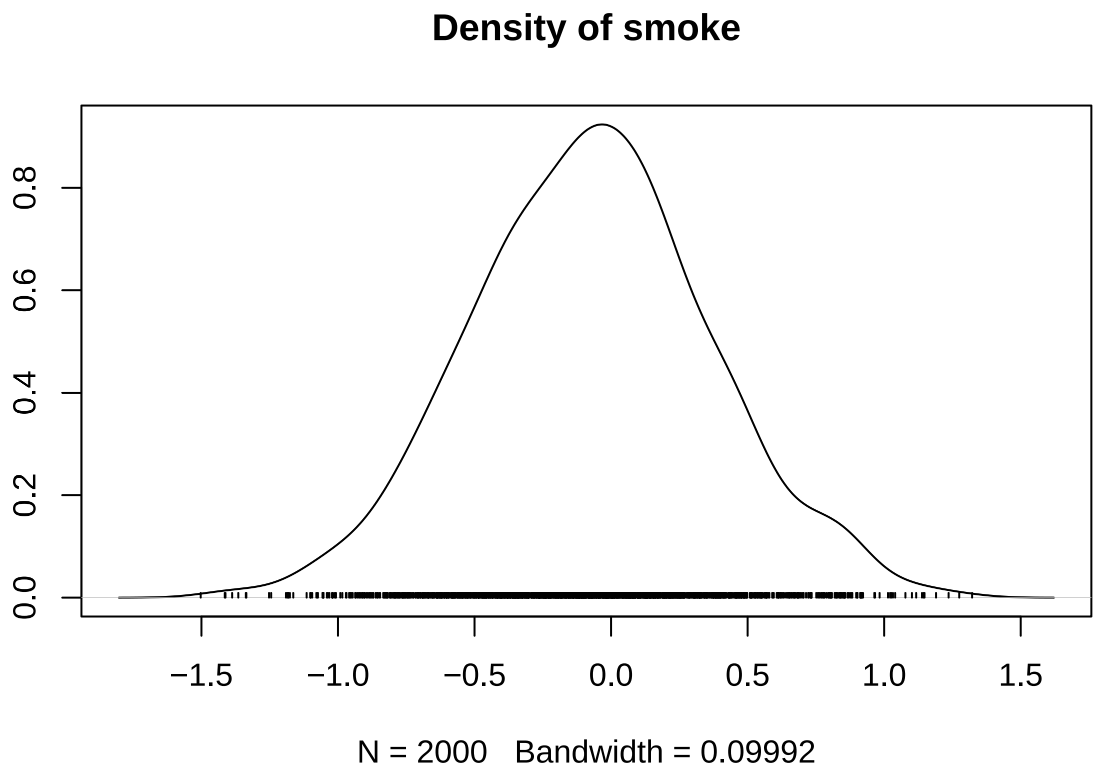

Density of (Intercept)
0.00
0.01
0.02
0.03
0.04
0.05
60
70
80
90
100
110
120
N = 2000 Bandwidth = 1.824

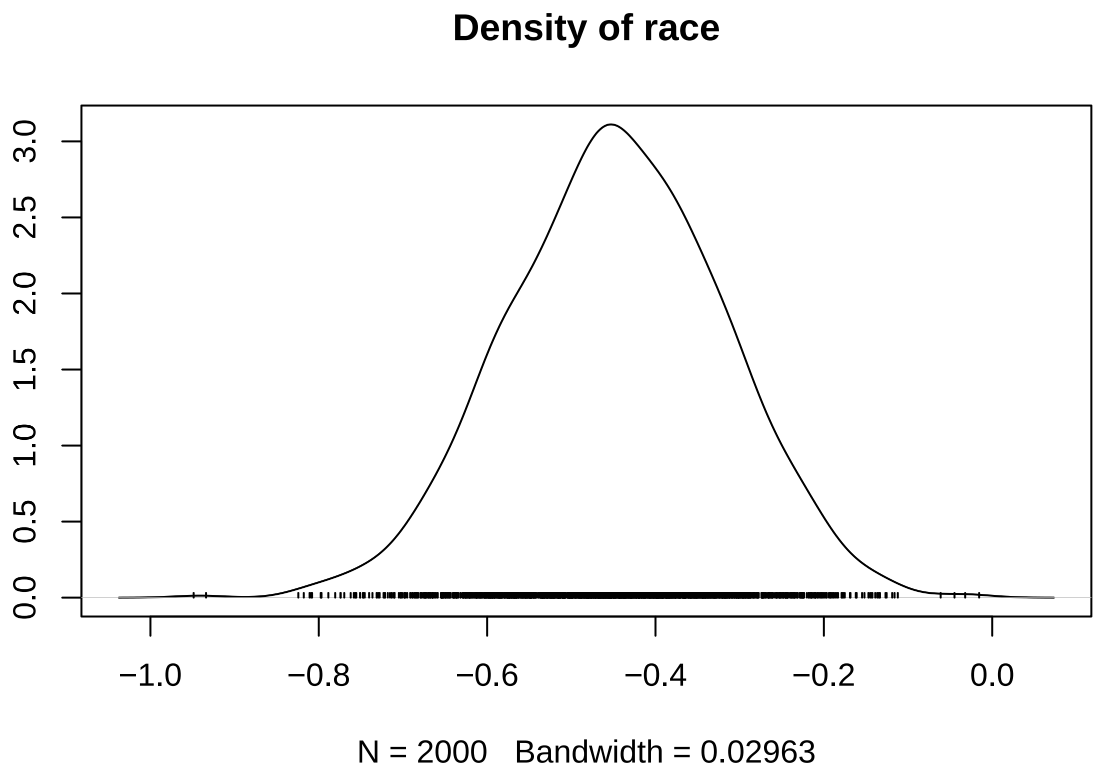

Density of gestation
0
10
20
30
40
50
−0.01
0.00
0.01
0.02
0.03
N = 2000   Bandwidth = 0.001578

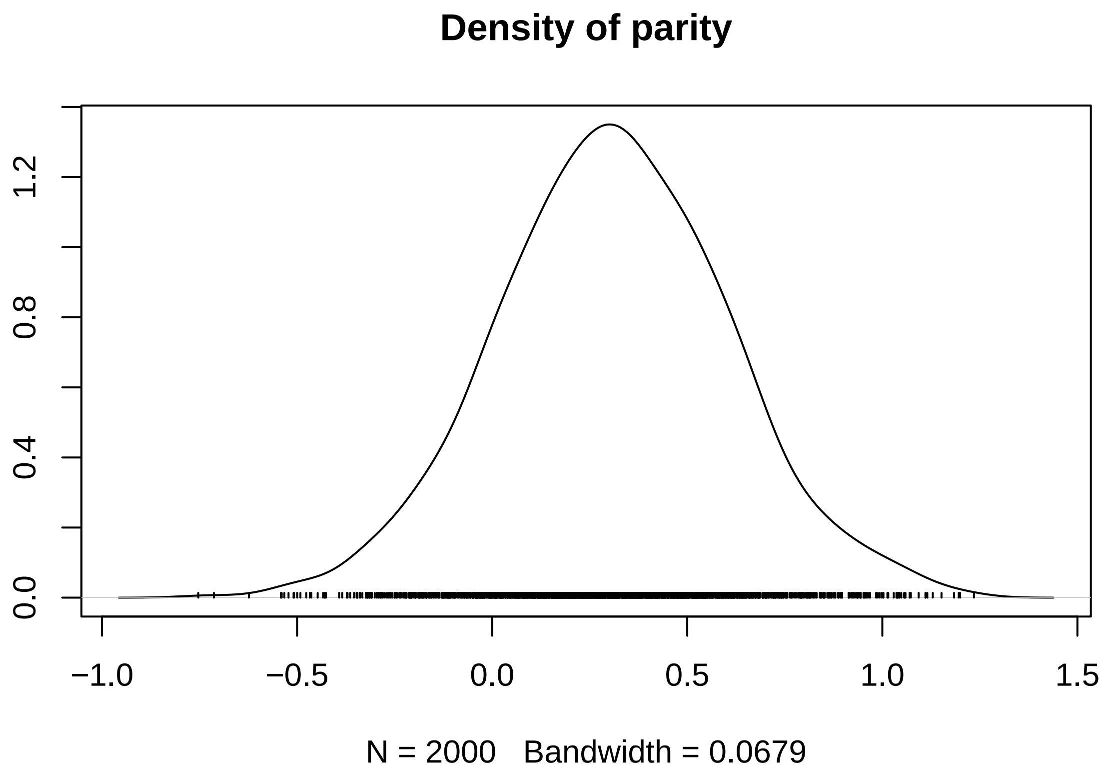

Density of age
0
1
2
3
4
−0.2
0.0
0.2
0.4
N = 2000   Bandwidth = 0.02104

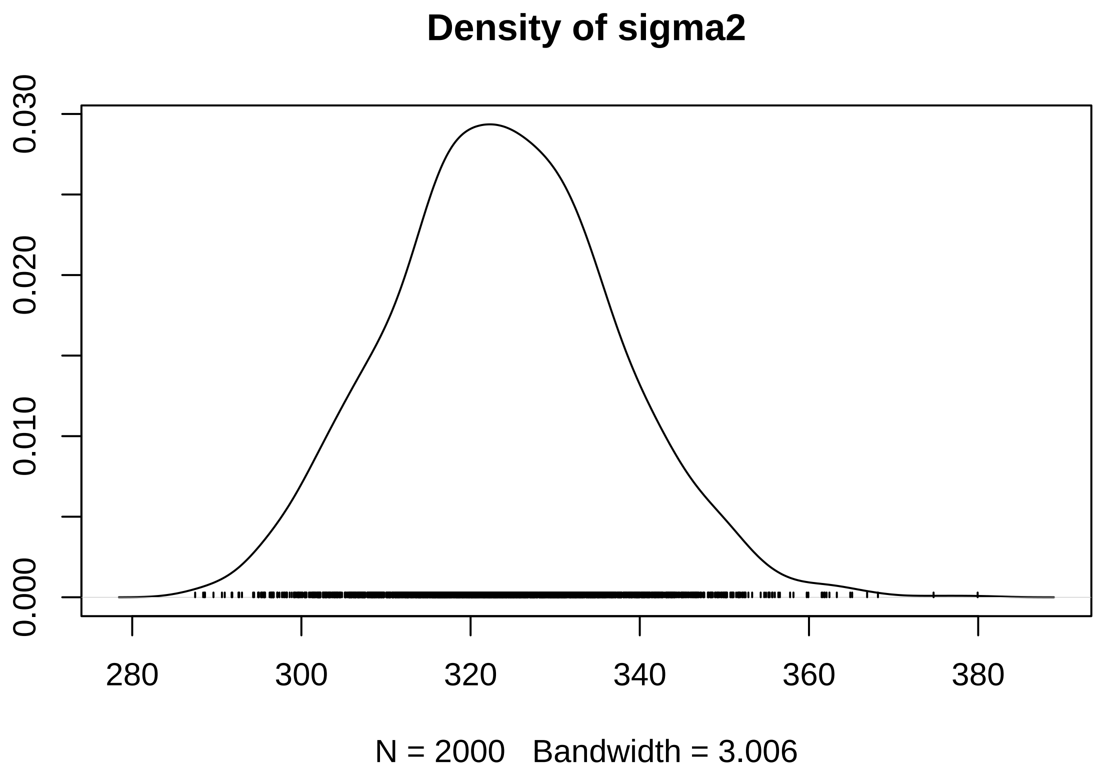

Density of ht
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0.0
0.2
0.4
0.6
0.8
N = 2000 Bandwidth = 0.02837


Density of wt1
0
20
40
60
80
−0.02
−0.01
0.00
0.01
N = 2000 Bandwidth = 0.00101

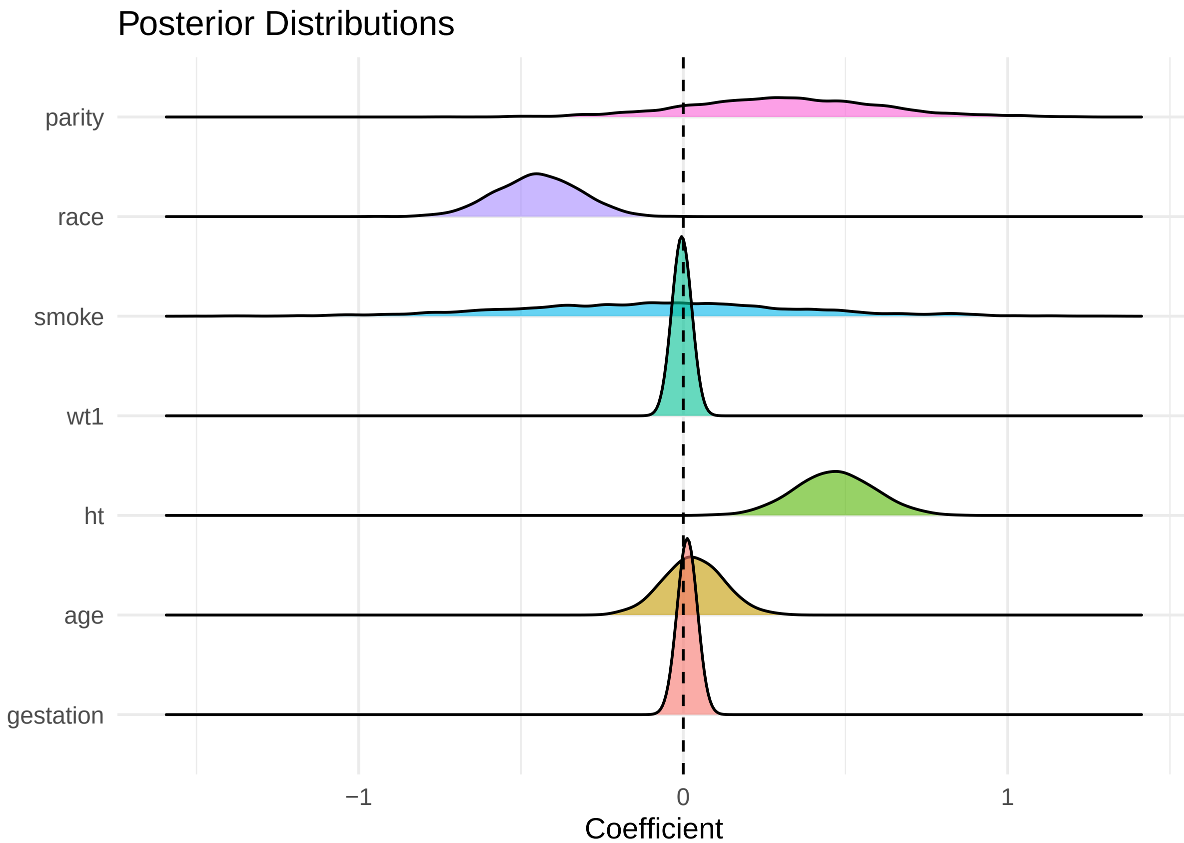

Density of smoke
0.0
0.2
0.4
0.6
0.8
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
N = 2000 Bandwidth = 0.09992


Density of race
0.0
0.5
1.0
1.5
2.0
2.5
3.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
N = 2000   Bandwidth = 0.02963


Density of parity
0.0
0.4
0.8
1.2
−1.0
−0.5
0.0
0.5
1.0
1.5
N = 2000 Bandwidth = 0.0679




```r
#ridgeline plot
m <- as.matrix(model)

df <- data.frame(
  gestation = m[, "gestation"],
  age = m[, "age"],
  ht = m[, "ht"],
  wt1 = m[, "wt1"],
  smoke = m[, "smoke"],
  race = m[, "race"],
  parity = m[, "parity"]
)

df_long <- melt(df)

ggplot(df_long, aes(x = value, y = variable, fill = variable)) +
  geom_density_ridges(alpha = 0.6) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(title = "Posterior Distributions",
       x = "Coefficient", y = "") +
  theme_minimal() +
  theme(legend.position = "none")
```



Posterior Distributions
parity
race
smoke
wt1
ht
age
gestation
−1
0
1
Coefficient